White Paper

OpenAlex as the Literature Database of Choice

true

today

Abstract

The short abstract goes here and can go over multiple lines. ...

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{r} #| label: setup #| include: false # This is needed to execute
the inline code
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Reviewed by:

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Introduction

OpenAlex is a scholary data source as for example Web of Science or Scopus.

Advantges OpenAlex in comparison to other databases

The comparison needs to be done inseveral aspects which are:

- available works and number of OpenAccess works
- global coverage (i.e. Global South)
- openness of data and code used
- Web Access and API quality
- costs

We will go throuh these points in more detail now.

Available Works and Number of OpenAccess works

	Number of works	Open Access Works	Citations	Price	Data Openness	Org Structure
OpenAlex	243M	48M	1.9B	Freemium	Fully open, CC0 license	Non-profit
Scopus	87M	20.5M (ref)	1.8B	Subscription	Closed	For Profit
Web of Science (core)	87M (ref)	12M (ref)	1.8B	Subscription	Closed	For Profit
Dimensions	135M	29M (ref)	1.7B	Freemium	Partly open, personal use	For Profit
Google Scholar	389M (estimated)	?	?	Free	Closed	For Profit
Crossref	145M	20M	1.45B	Free	Fully open, CC0 license	Non-profit

(From: OpenAlex)

Global Coverage (i.e. Global South)

OpenAlex has a broad coverage for articles and authors from the Global South, which is not the case for other databases REF???.

Openness of Data and Code used

OpenAlex is free and open, i.e. it is free to use for everybody without any subscription or registration, by design. The data is freely available (database dumps are possible and easy to do) and the code used is also openly available.

Web Access and API quality

OpenAlex has, as the other scholary data sources, a web interface. It is still in beta phase, so elements can change and aspects will improve over time, but it is already very usable and based on personal preferences easier to use than WoS or Scopus.

API access is important for accessing the database programmatically. An example is a search for all articles covering a specific topic. Usually, this is an iterative

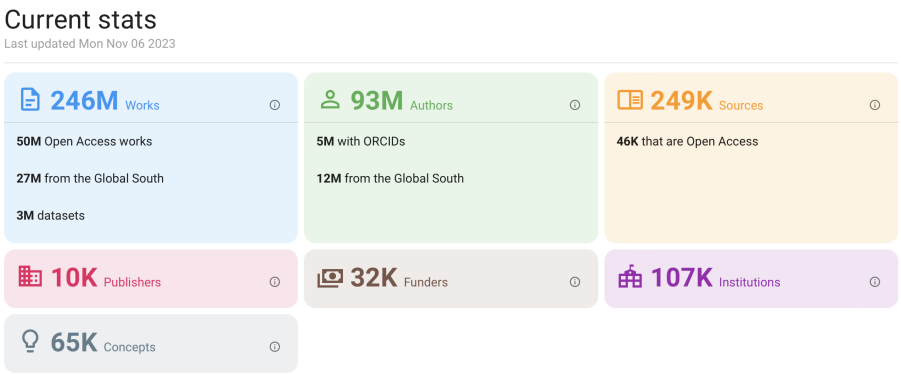

Figure 1: From: OpenAlex

provess which needs refinement of the search terms. THe traditional way is to type the search term in the search box and at east writew down the number of hits, change the search term and repeat the process. especially when other comparisons should also be done (e.g. assessment of cerrtain key-papers are in the search results). This is a very tedious process and can be automated using the API. In this case, the search is defined e.g. in R or any other programming language and the search is done automatically. The results are then analysed and the search terms are refined and the search is repeated, and the results can be shown in a clear, concise and reproducible report.

In contrast to the APIs assessed, (WoS, Scopus), the [OpenAlex API(https://docs.openalex.org)] is relatively simple but provides all the neccessary functionality needed for IPBES and for all other tasks.

To simplify the API usage even more, the excellent R package openalexR is available to access the API from R.

Costs

OpenAlex is free to use, which is not the case for most other databases. If the limit for free usage (100k API calls per day with a maximum of 10 per second) is not enaough, a Premium subscription is offered. The pricing is determined on a case by case basis.

For the use cases of IPBES, the free tier is more than enough, especially as the API restrictions are email based and multiple users can use their own email adresses..

The API is, as is the access through the website, free for OpenAlex.

In contraast, the API costs for WoS are getting into the several thousands per dollars per year. In addition to the costs,

Conclusion

@article{Michael2023, title = {A Systematic Evidence Map of Intervention Evaluations to Reduce Gang-Related Violence}, author = {Michael, Richardson and Mark, Newman and Gregory, K. Berry and Claire, Stansfield and April, Coombe and James, Hodgkinson}, year = {2023}, month = jun, journal = {Journal of Experimental Criminology}, pages = {undefined}, publisher = {{Springer Science+Business Media}}, issn = {1572-8315, 1572-8315}, doi = {10.1007/s11292-023-09574-w}, langid = {english}, keywords = {/unread}, file = {/Users/rainerkrug/Library/Mobile Documents/com$_{\text{apple}}$CloudDocs/Zotero Attachments/2023/Michael2023___Michael et al_2023_A systematic evidence map of intervention evaluations to reduce gang-related.pdf} }